

Anant Sharma

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Education

M.Sc. - Quantum Technologies & Engineering (QuanTEEM)

UBFC - Université Bourgogne-Franche-Comté, Dijon, France (Semester 1 & 2) 2024 - 2025

Aarhus University, Aarhus, Denmark (Semester 3) 2025 - 2026

Thesis Title: “Effects of spurious modes on external loss rates of superconducting resonators”

Advisor: Dr. Yuki Nojiri

B.Sc. - Physics

Fergusson College, Pune, India 2020 - 2023

Grade: A (8.38/10)

Thesis Title: “Blind Quantum Computing”

Advisor: Dr. Alka V. Deshmukh & Mr. Rikteem Bhowmick

Awards & Posters

Noise Characterization of Flux-Tunable Resonators [†]	UBFC M1 Project Presentation June - 2025
Erasmus Mundus Scholar (2-year full-ride scholarship)	Dijon, France. September - 2024
Hide & Seek: Blind Quantum Computing (Third Prize) [†]	Frontier In Physics XVI, Pune. April - 2023
Cosmic Web: Sloan Digital Sky Survey [†]	Poster Exhibition Competition, Pune. January - 2023
Protein Folding: The Quantum Style [†]	Avishkar - Zonal Level Candidate, Pune. September - 2022
Krishi Help (Winner of AgriTech Stream)	FC Hackathon, Pune. March - 2022
Team Neutrino: Baikal Telescope (First Prize) [†]	Poster Exhibition Competition, Pune. December - 2021
BMW Quantum Computing Challenges, Stage - 2	Q2B Conference 2021. October - 2021

Research & Internship Experiences

Effects of spurious modes on external loss rates of superconducting resonators February 2026 - August 2026

Alice & Bob, Paris, France

Advisor: Dr. Yuki Nojiri

- Developed models and design-automation tools to analyze superconducting circuits and validate experimental measurements.
- Built comprehensive benchmarking pipelines and fitting methods to characterize device performance and explain discrepancies between simulations and measurements.
- Maintained research code and technical documentation while collaborating with chip-design, simulation, and experimental teams.

Simulation of Ba⁺ Ions qubit coherence time September 2025 - January 2026

Aarhus University, Aarhus, DK

Advisor: Dr. Michael Drewsen

- Modeled Rabi oscillations and decoherence mechanisms (detuning jitter, Rabi-frequency noise) in a driven two-level Ba⁺ ion system by numerically solving the optical Bloch equations in Python.
- Implemented dressed-atom and double-dressed-state formalisms to demonstrate noise suppression via continuous dynamical decoupling and phase-modulated driving fields.
- Validated theoretical predictions against experimental Rabi scan data from the Aarhus University Ion Trap Lab, confirming extended coherence times under phase-modulated driving

Noise Characterization of Flux Tunable Resonator May 2025 - August 2025

Chalmers University of Technology, Gothenburg, SE

Advisor: Dr. Witlef Wiczorek

- Guided by PhD student Achintya Paradkar, worked with superconducting circuits and developed scripts to acquire data and post-process a flux-tunable resonator (FTR)
- Assembled a Homodyne Readout setup for noise readout of the FTRs. Experience with operating dilution fridges and experimental techniques.

An Efficient Quantum algorithm for Travelling Salesman Problem July 2023 - July 2024

Centre for Quantum Engineering, Research & Education (CQuERE), TCG - CREST, Kolkata, IN

[†]Posters presentations

Advisor: Dr. Shibdas Roy

- Developed a novel quantum algorithm to realise a *conditional* exponential speed-up for the travelling salesman problem. (Hamiltonian Cycle Problem) solving faster than the state-of-the-art Grover's algorithm. Using techniques such as Quantum Phase Estimation (QPE), and Quantum Fourier Transformation (QFT).
- Development of robust circuit logic verification protocols, analysis of quantum circuit and run-time complexities.

Universal Blind Quantum Computing (UBQC) Simulator

August 2022 - June 2023

Qulabs - Quantum Research Intern (Bachelors Thesis)

Advisor: Mr. Aman Gupta, Mr. Rikteem Bhowmick

- Developed a UBQC simulator using qiskit, measurement-based quantum computing (MBQC) protocol and cluster states.
- Benchmarked the simulator for quantum gates and common quantum algorithms. Explored techniques like circuit-knitting-toolbox, OPENQASM3.0 and Dynamic Circuits for the real backend implementation of the simulator.

BMW Quantum Computing Challenges - 2021

August 2021 - January 2022

BosonQpsi - Quantum Developer Intern

Advisor: Mr. Rut Lineswala & Mr. Abhigyan Mishra

- Suggested an annealer-based approach for the solution of the linear system using a Quadratic Unconstrained Binary Optimisation (QUBO) formulation. Source
- Extensively benchmarked the D-Wave annealers to solve up to 3100 x 3100 sparse matrices and documented the run time, minimum energy, solution, and error in the solution. Learnt about the complexity of the code, and worked out different optimised encodings that used fewer resources for the same matrix size, allowing us to benchmark larger matrices.

Skills Summary

- **Coding Languages:** Python, C++, MERN Stack
- **Quantum Packages:** Qiskit, Cirq, Ocean - SDK, Classiq, Qua
- **Tools:** VNA, Spectrum Analyzer, OPX+, Octave, Dilution Fridges, RF Measurements, KiCAD, Astrometrica, FEKO, GIT
- **Languages:** English, Hindi, Marathi
- **Soft Skills:** Leadership, Management, Communication, Public Speaking

Summer Schools

European Quantum Technology Summer School

Paderborn, DE, June 2025

Selected for a fully funded week-long summer school on quantum technologies. Covering multiple domains of quantum technology as lectures, workshops, and lab tours.

Carnot Workshop on Quantum Thermodynamics and Open Quantum Systems

Dijon, FR, November 2024

A series of lectures delivered on Quantum thermodynamics and open quantum systems by various researchers in the domain.

Equinox - Cohort 5

April 2022 - May 2022

Radio Antenna Simulations

Learning about various radio antennae their simulation and advantages. Analyzing the antenna for low and high band frequencies and comparing their performance. Determined various parameters of an antenna performing parameter sweeps and further optimizing. Used Software: **PostFEKO, CADFEKO**. Used Antenna Types: **Bowties**

Publications

Journal Publications

1. **Sharma, A.**, Deshpande, N., Ghosh, S., Das, S. & Roy, S. An efficient quantum algorithm for the traveling salesman problem. Quantum Information & Computation 25, 71–81 (2025). (Link)

Book Chapter

1. **Sharma, A.**, Paradkar, A., Rao, V.N. (2022). Quantum Technologies II: Cryptography, Blockchains, and Sensing. In Quantum and Blockchain for Modern Computing Systems: Vision and Advancements pp [55 - 102].

Outreach & Volunteering

Astroclub FC

September 2020 - April 2023

Fergusson College

- **Core Team (2021 - 2023)** Managing & organizing Astroclubs' flagship events Frontiers in Physics (Edition XVI) viz. A 2-day national level seminar & The annual poster exhibition competition with over 25 volunteers, eminent researchers and 3000 + visitors.

- **Volunteer (2020 - 2021)** Volunteering in activities like scientific outreach (lectures in rural areas on basic astronomy) on the occasion of National Science Day (February 28th), assisting telescope handling sessions and poster exhibitions competitions.

Skills Leadership, Public Speaking, Management, Scientific Outreach, Content creation & writing

QIndia

Jan 2021 - Jan 2022

QWorld

- Started working as a content writer for QIndia, a vertical of QWorld, creating content focused on quantum computing and information. Additionally, played a role in the web development team by contributing to the creation of the QIndia website. Engaged in video editing and event organization as well.

Relevant Courses

Quantum Mechanics - II, III

Universite Bourgogne Europe, Aarhus University

Quantum Information and Engineering

Universite Bourgogne Europe

Quantum Thermodynamics and Control Theory

Universite Bourgogne Europe

Platforms for Quantum Technology

Aarhus University

Linear Algebra, Real Analysis, Vector Calculus

Fergusson College (Dept. of Mathematics)

Statistical Physics, Lasers, Mathematical Methods in Physics,

Solid State Physics, Atomic and Molecular Physics, Quantum Mechanics-I,

Classical Electrodynamics

Fergusson College

References

Dr. Yuki Nojiri

Alice & Bob, Paris, FR

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Dr. Shibdas Roy

Center for Quantum Engineering, Research and Education (CQuERE), Kolkata, IN

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Dr. Raka V. Dhabada

Fergusson College, Pune, IN

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